INSIGHT MINDS

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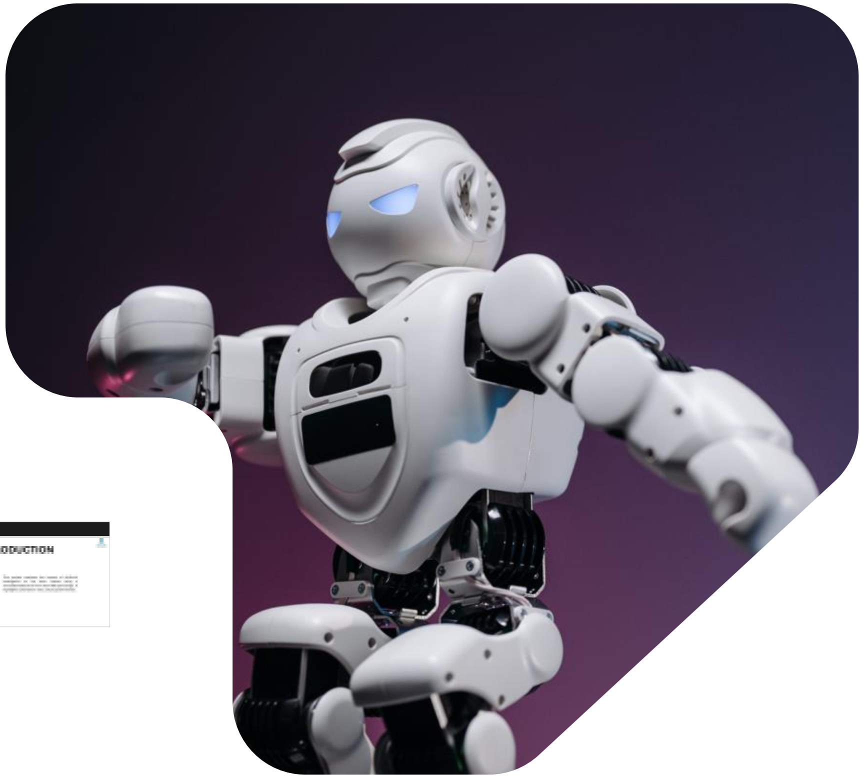

AI-Powered Job Market Insights

Done by:

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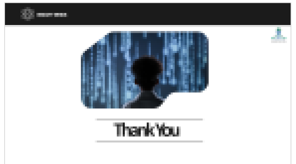
Supervised by: DR. Amal Mahmoud



INTRODUCTION

This project analyzes the impact of Artificial Intelligence on the labor market using a simulated dataset of more than 500 job listings. It highlights automation risks, future growth trends.

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OBJECTIVES

TRENDS

Explore employment trends influenced by AI.

JOBS

Identify vulnerable job categories at risk of automation.

INSIGHTS

Provide insights for policymakers, companies, and professionals.

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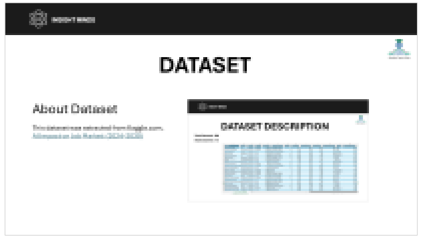
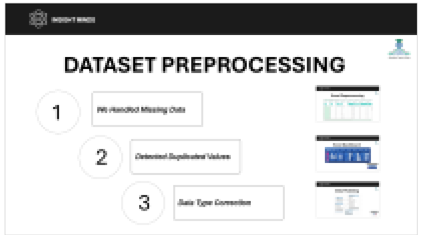
METHODOLOGY



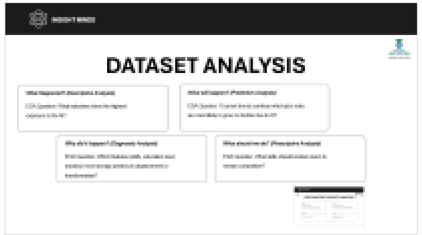
Dataset

- About Dataset
- Dataset Description



Data Preprocessing

- Data Preparation.
- Data Cleaning & Preprocessing.
- Data modeling



Data Analysis

- EDA
- Data visualization

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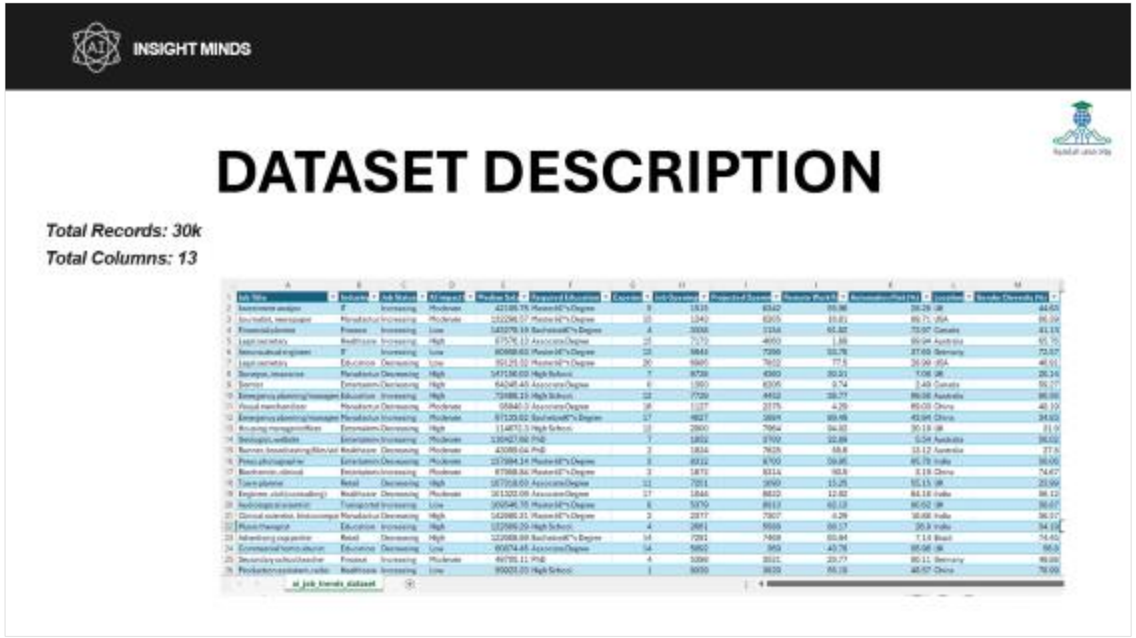
DATASET

About Dataset

This dataset was extracted from Kaggle.com,
[AI Impact on Job Market: (2024–2030)]



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DATASET DESCRIPTION

Total Records: 30k

Total Columns: 13

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Job Title	Industry	Job Status	AI Impact I	Median Sala	Required Education	Experier	Job Opening	Projected Openin	Remote Work Ra	Automation Risk (%)	Location	Gender Diversity (%)
2	Investment analyst	IT	Increasing	Moderate	42109.76	Masterâ€™s Degree	5	1515	6342	55.96	28.28	UK	44.63
3	Journalist, newspaper	Manufactur	Increasing	Moderate	132298.57	Masterâ€™s Degree	15	1243	6205	16.81	89.71	USA	66.39
4	Financial planner	Finance	Increasing	Low	143279.19	Bachelorâ€™s Degree	4	3338	1154	91.82	72.97	Canada	41.13
5	Legal secretary	Healthcare	Increasing	High	97576.13	Associate Degree	15	7173	4060	1.89	99.94	Australia	65.76
6	Aeronautical engineer	IT	Increasing	Low	60956.63	Masterâ€™s Degree	13	5944	7396	53.76	37.65	Germany	72.57
7	Legal secretary	Education	Decreasing	Low	39123.32	Masterâ€™s Degree	20	6985	7832	77.5	28.99	USA	46.91
8	Surveyor, insurance	Manufactur	Decreasing	High	147150.03	High School	7	9738	4360	30.51	7.08	UK	20.14
9	Dentist	Entertainm	Decreasing	High	64245.48	Associate Degree	0	1393	8206	9.74	2.49	Canada	59.27
10	Emergency planning/managen	Education	Increasing	High	72488.15	High School	12	7729	4452	38.77	99.56	Australia	60.56
11	Visual merchandiser	Manufactur	Decreasing	Moderate	95840.3	Associate Degree	16	1127	2375	4.29	69.03	China	48.19
12	Emergency planning/managen	Manufactur	Increasing	Moderate	97133.62	Bachelorâ€™s Degree	17	4627	1684	89.49	43.94	China	34.83
13	Housing manager/officer	Entertainm	Decreasing	High	114672.3	High School	12	2800	7964	94.02	20.19	UK	31.9
14	Geologist, wellsite	Entertainm	Increasing	Moderate	116427.68	PhD	7	1852	3700	32.89	5.54	Australia	56.02
15	Runner, broadcasting/film/vid	Healthcare	Decreasing	Moderate	43059.04	PhD	2	1824	7625	55.6	13.12	Australia	27.5
16	Press photographer	Entertainm	Decreasing	Moderate	137894.14	Masterâ€™s Degree	3	8312	9700	59.95	95.76	India	56.06
17	Biochemist, clinical	Entertainm	Increasing	Moderate	67908.84	Masterâ€™s Degree	2	1872	5314	93.9	3.19	China	74.67
18	Town planner	Retail	Decreasing	High	107318.63	Associate Degree	11	7251	1090	15.25	55.15	UK	23.99
19	Engineer, civil (consulting)	Healthcare	Decreasing	Moderate	101322.09	Associate Degree	17	1044	8832	12.92	64.18	India	56.12
20	Audiological scientist	Transportat	Increasing	Low	109546.76	Masterâ€™s Degree	8	5379	8913	82.13	80.62	UK	38.87
21	Clinical scientist, histocompat	Manufactur	Decreasing	High	142660.31	Masterâ€™s Degree	2	2377	7307	4.29	16.88	India	56.57
22	Music therapist	Education	Increasing	High	122569.29	High School	4	2661	5588	88.17	26.8	India	34.19
23	Advertising copywriter	Retail	Decreasing	High	122668.89	Bachelorâ€™s Degree	14	7291	7469	65.94	7.14	Brazil	74.45
24	Commercial horticulturist	Education	Decreasing	Low	60874.45	Associate Degree	14	5892	369	43.78	95.96	UK	56.8
25	Secondary school teacher	Finance	Increasing	Moderate	49705.11	PhD	4	5398	3531	20.77	60.11	Germany	49.88
26	Production assistant, radio	Healthcare	Increasing	Low	59923.23	High School	1	8039	3829	55.19	48.57	China	78.99

ai_job_trends_dataset

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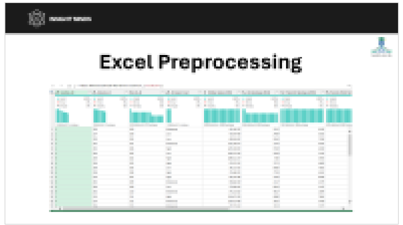
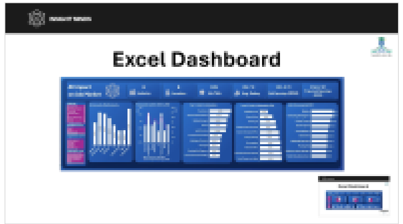
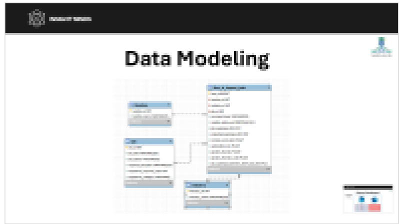
DATASET PREPROCESSING

1. *We Handled Missing Data*
2. *Detected Duplicated Values*
3. *Data Type Correction*







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Excel Preprocessing

```
= Table.RemoveColumns(#"Reordered Columns2",{"Location"})
```

	location_id	industry_id	job_id	AI Impact Level	Median Salary (USD)	Job Openings (2024)	Projected Openings (2030)	Remote Work Rati
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty
	8 distinct, 0 unique	8 distinct, 0 unique	828 distinct, 670 unique	3 distinct, 0 unique	1000 distinct, 1000 unique	948 distinct, 898 unique	953 distinct, 906 unique	945 distinct, 890 uniqu
1	1	200	200	Moderate	42,109.76	1515	6342	
2	1	214	200	Low	41,997.68	2409	5015	
3	5	200	204	Low	60,956.63	5944	7396	
4	2	202	201	Moderate	132,298.57	1243	6205	
5	1	202	206	High	147,150.03	9738	4360	
6	3	204	202	Low	143,279.19	3338	1154	
7	6	200	202	High	109,421.37	985	3954	
8	4	206	203	High	97,576.13	7173	4060	
9	2	208	205	Low	39,123.32	6985	7832	
10	4	208	208	High	72,488.15	7729	4452	
11	3	210	207	High	64,245.48	1393	8206	
12	6	202	209	Moderate	95,840.30	1127	2375	
13	2	206	209	Low	37,084.34	9919	1155	
14	6	202	210	Moderate	97,133.62	4627	1684	
15	1	210	211	High	114,672.30	2800	7964	
16	4	210	212	Moderate	116,427.68	1852	3700	
17	4	206	213	Moderate	43,059.04	1824	7625	

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Excel Dashboard

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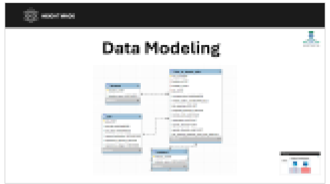
Excel Dashboard



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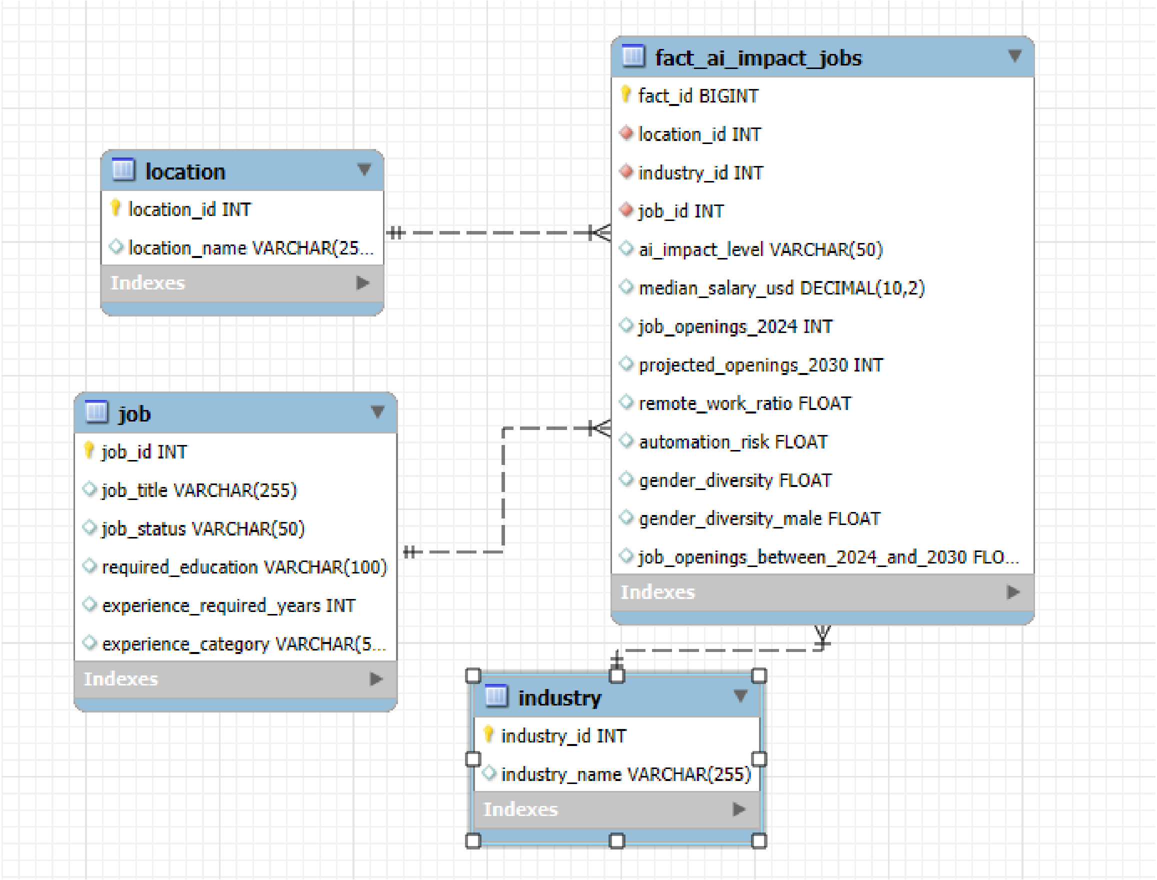
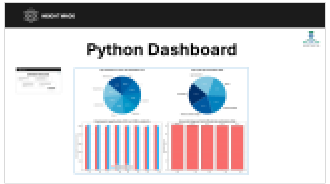
Data Modeling





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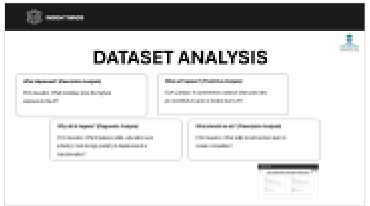
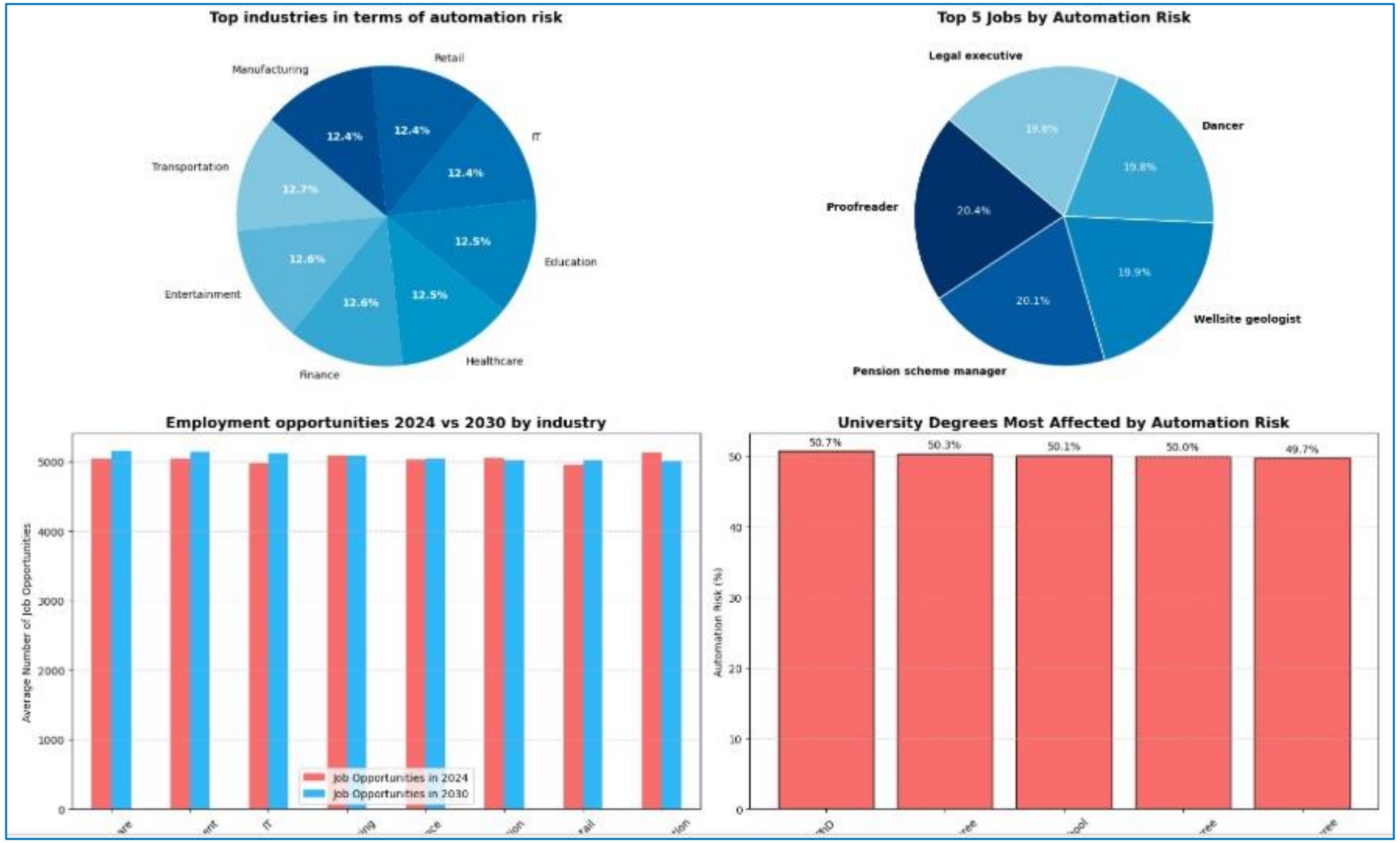
Python Dashboard





DATASET ANALYSIS

What Happened? (Descriptive Analysis)

EDA Question: What industries show the highest exposure to the AI?

What will happen? (Predictive Analysis)

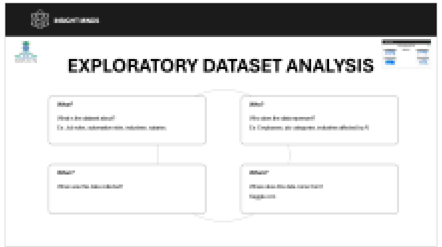
EDA Question: If current trends continue which jobs roles are most likely to grow or decline due to AI?

Why did it happen? (Diagnostic Analysis)

EDA Question: Which features (skills, education level, industry) most strongly predict job displacement or transformation?

What should we do? (Prescriptive Analysis)

EDA Question: What skills should workers learn to remain competitive?



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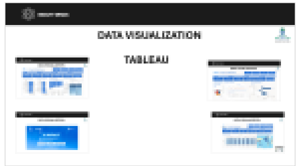

EXPLORATORY DATASET ANALYSIS

What?

What is the dataset about?

Ex: Job roles, automation risks, industries, salaries.

Who?

Who does the data represent?

Ex: Employees, job categories, industries affected by AI

When?

When was this data collected?

Where?

Where does this data come from?

Kaggle.com

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DATA VISUALIZATION

TABLEAU

DATA VISUALIZATION

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DATA VISUALIZATION

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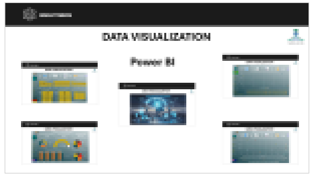

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DATA VISUALIZATION

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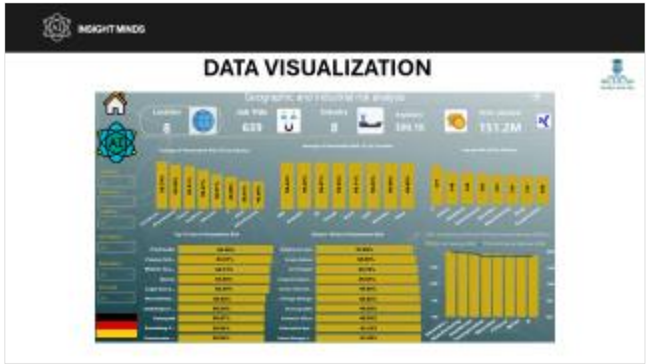
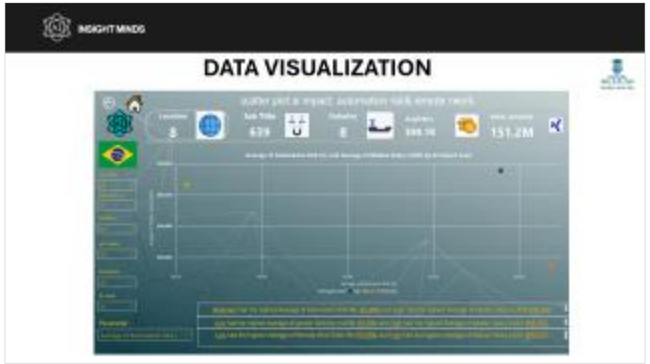
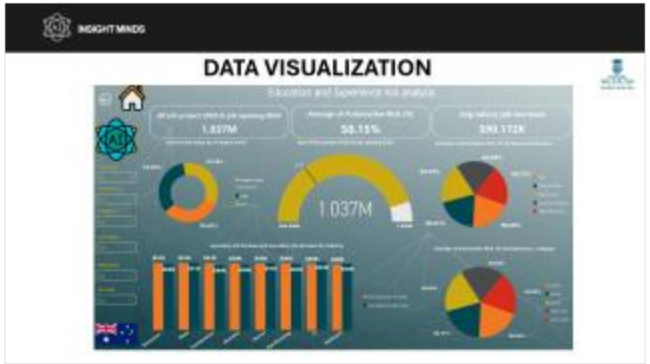
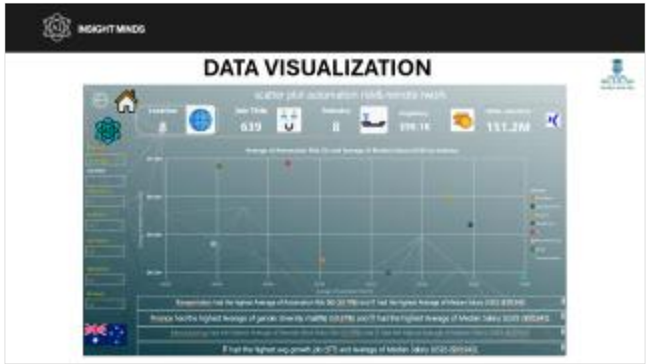
Power BI









DATA VISUALIZATION

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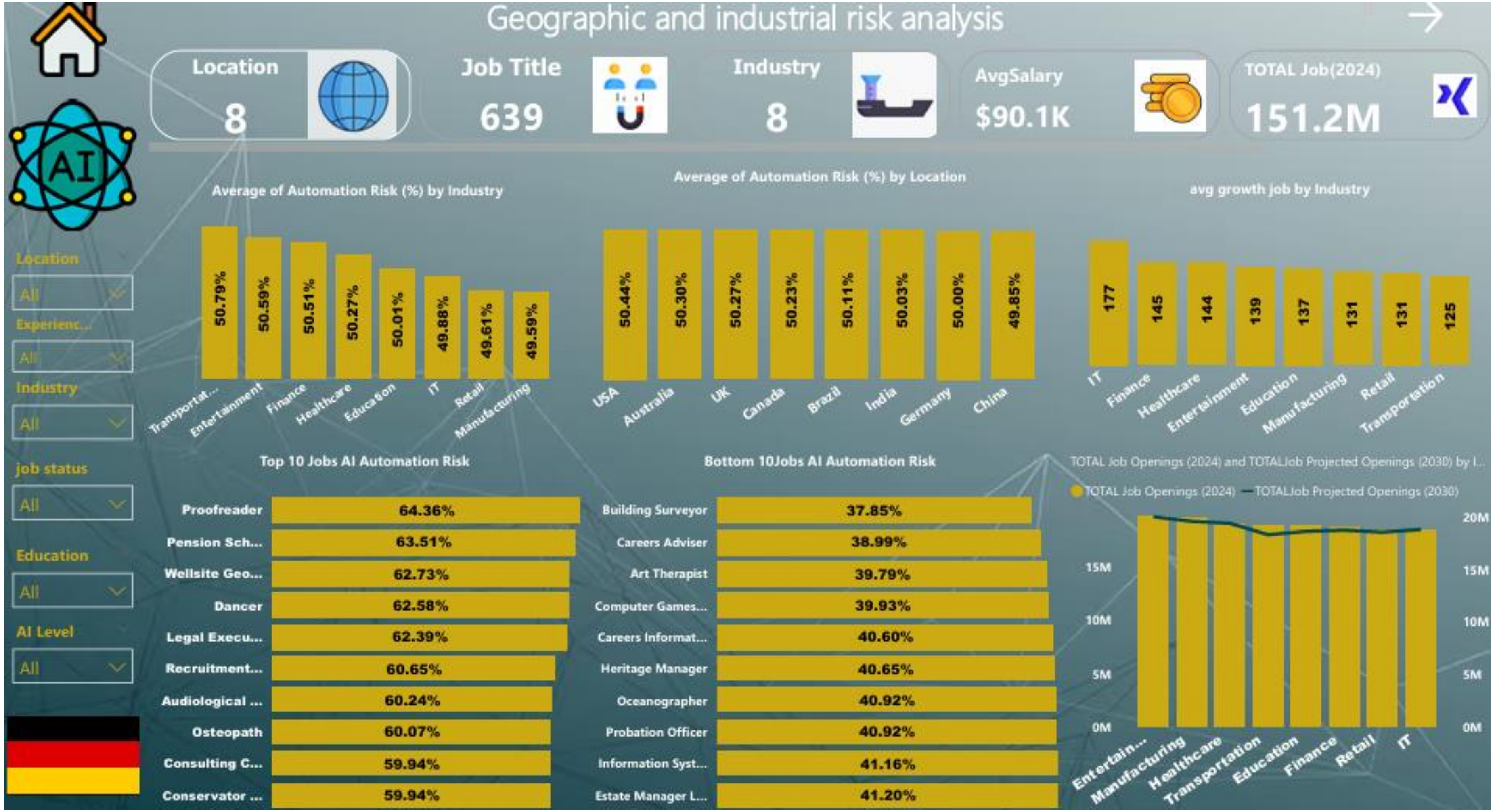

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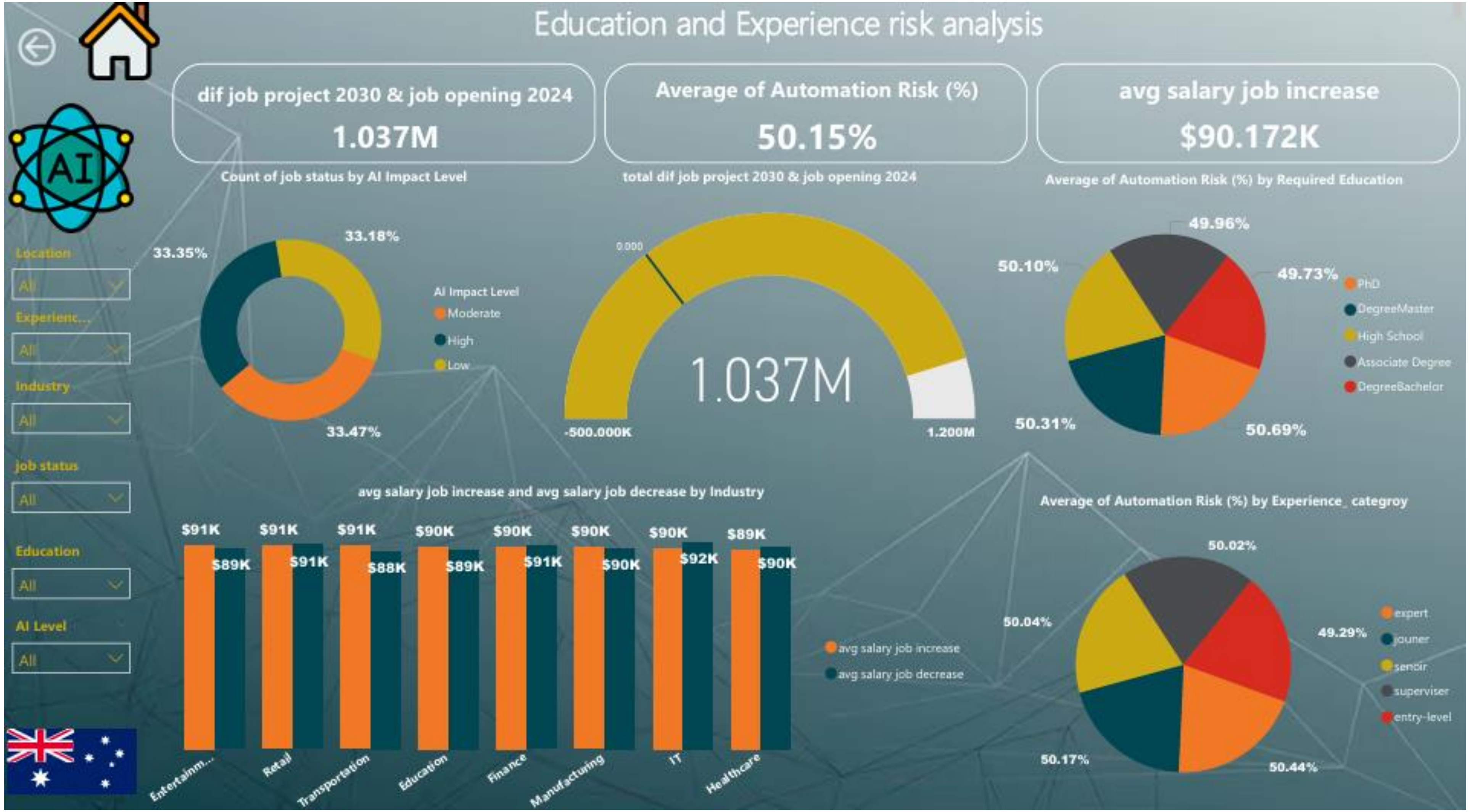

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DATA VISUALIZATION

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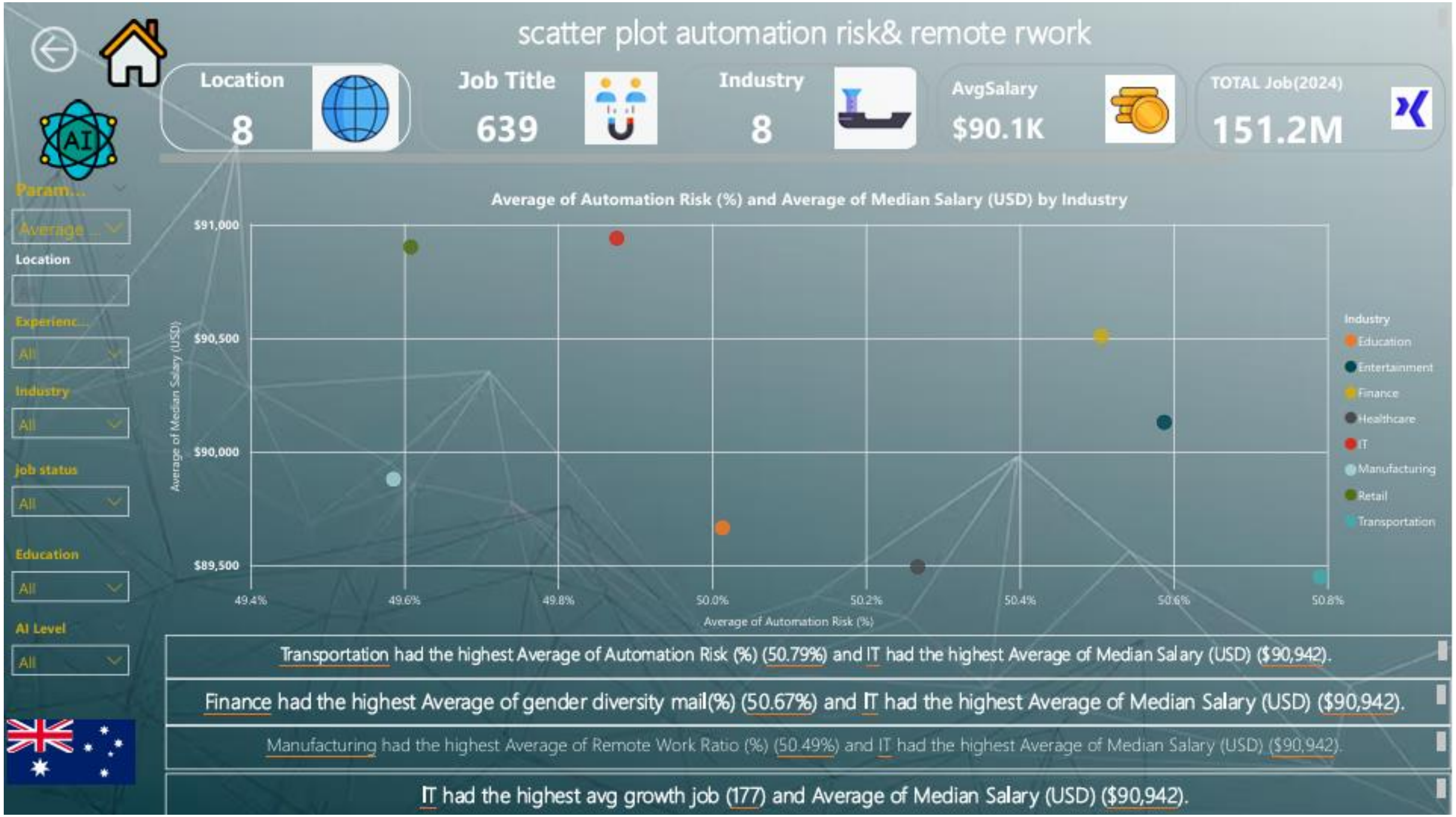

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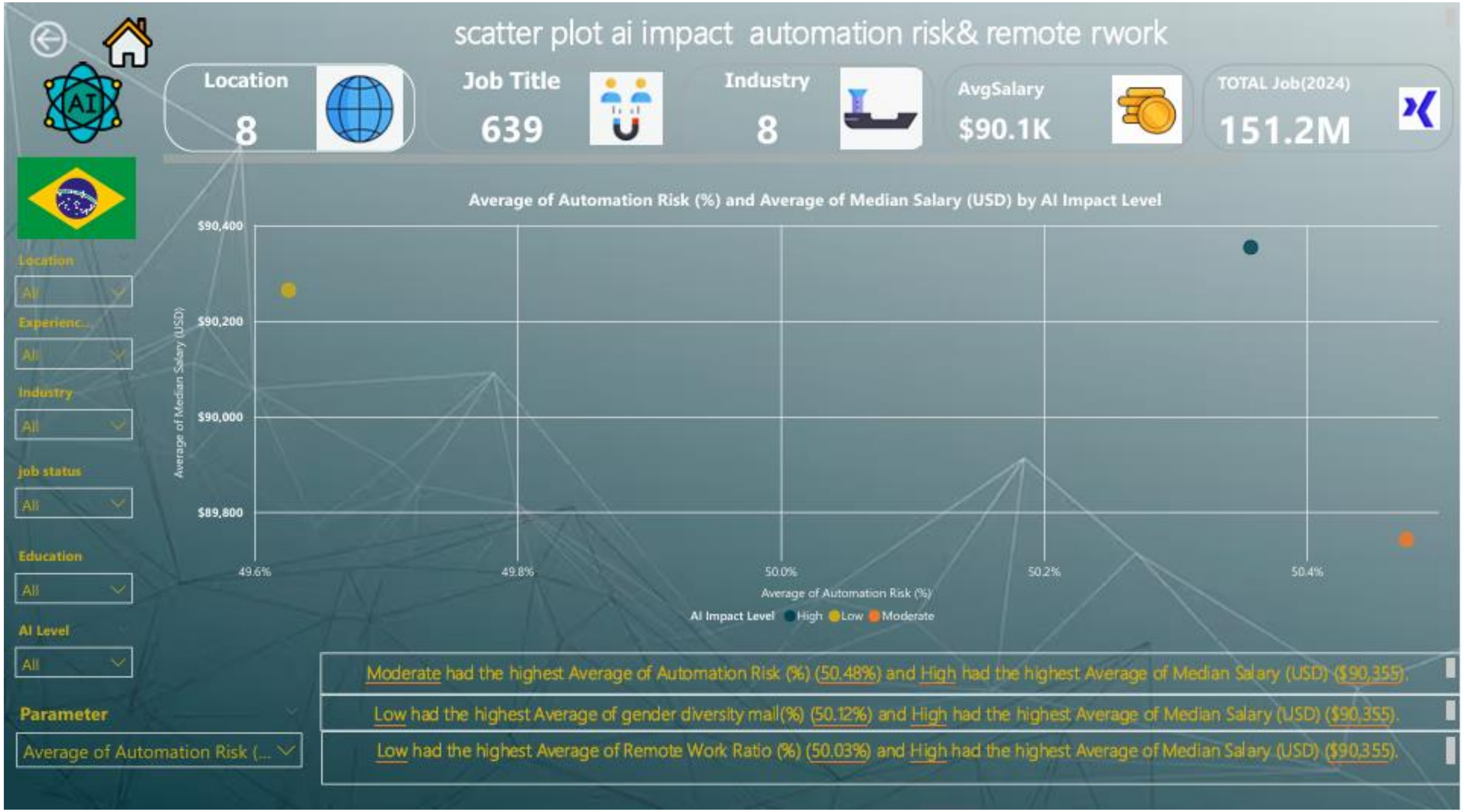

Conclusion

The analysis demonstrates that AI is reshaping the global job market through a balanced mix of automation, job shifts, and skill evolution. While certain job categories—particularly routine and operational roles—face high automation risk, the market remains stable overall, with projected openings showing only moderate change through 2030. Growth is concentrated in expert, supervisory, and high-skilled technical roles, reinforcing the increasing value of specialized expertise.

Geographically, automation exposure is consistent across countries, indicating that AI's impact is universal rather than region-specific. Industries such as entertainment, healthcare, education, and manufacturing are expected to experience steady growth, supported by evolving skill demands and technological integration.

Overall, the findings highlight the need for continuous reskilling, stronger digital capabilities, and proactive market adaptation to ensure workforce readiness for the coming decade.

Main Outcomes

1. AI Automation Risk
The job market shows **balanced automation exposure**, with risk levels distributed almost evenly across Low (32.8%), Moderate (33.7%), and High (33.5%) categories.
Certain roles—such as **Proofreaders, Technical Brewers, Recruitment Consultants, and Sales Executives**—face automation risks above **70%**, indicating strong potential for AI-driven displacement.
In contrast, **creative, specialized, and interpersonal roles** (e.g., Art Therapists, Heritage Managers, Oceanographers) show significantly **lower automation risk**, mostly below **40%**.

2. Job Demand Forecast (2024–2030)
Total job openings are projected to grow from **151.19M in 2024** to **152.23M in 2030**, suggesting **stable but slow market expansion**.
Industries expected to grow most by 2030 include:

- **Entertainment** (+0.38M)
- **Manufacturing** (+0.02M)
- **Healthcare** (+0.31M)
- **Education & Finance** (moderate growth levels)

Main Outcomes

3. Jobs Decreasing by 2030
Several roles show noticeable decline in demand, including **Dietitians, Contracting Supervisors, Learning Mentors**, and **Manufacturing Specialists**, with job losses ranging from **58K to 84K**.
Declines are concentrated in roles with **routine, manual, or repetitive task structures**, reinforcing AI's impact on operational positions.

4. Experience-Level Outlook
Expert-level roles experience the largest growth (+583K), reflecting increased demand for innovation-driven and decision-making positions.
Junior-level roles decline significantly (-134K), highlighting automation's impact on entry-level and repetitive functions.
Supervisor and senior positions show moderate increases.

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Main Outcomes

5. Geographic Insights
Automation risk is relatively consistent globally, ranging between **49% and 51%**, indicating AI's **universal market impact**.
Countries like **Canada, Brazil, Germany, China, and Australia** display similar adoption and automation exposure rates.

6. Gender Diversity
Gender distribution remains **balanced across industries**, with male and female participation showing minimal variance, indicating that AI-driven shifts are **industry-based rather than gender-based**.

Recommendations

1. Prioritize Reskilling and Upskilling
Invest heavily in training related to **AI literacy, data analytics, automation tools, and digital workflows**.
Industries with high automation risk should shift toward **reskilling pathways** to preserve employment levels.

2. Support Transition for High-Risk Occupations
Develop targeted programs for roles exceeding **70% automation risk**, such as proofreading, sales, and administrative support.
Offer structured transition plans into **emerging roles** with lower automation exposure.

3. Strengthen Talent Pipelines for High-Growth Segments
Expand training for roles seeing high demand growth.
Encourage partnerships between governments, universities, and industries to align talent supply with market needs.

Recommendations

4. Enhance Workforce Planning Through Data
Organizations should use forecasting models to **predict skills shortages**, job shifts, and market gaps.
Cluster analysis and predictive analytics can support smarter workforce planning.

5. Leverage AI as an Enabler, Not a Replacement
Encourage companies to adopt AI to **augment** employee productivity rather than replace roles.
Emphasize hybrid workflows where AI supports repetitive tasks while humans focus on strategic, creative, and customer-facing work.

6. Prepare Policy and Regulatory Frameworks
Governments should introduce policies supporting safe, ethical, and transparent AI adoption.
Incentives can help industries modernize without accelerating unemployment.

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Thank You